CFAT 1.1.2

Five-number summaries and boxplots

1F, 1G · CAS and bound reference allowed

Name _____

/14

SECTION A · MULTIPLE CHOICE

Circle one answer. 6 marks

QUESTION 1 — 1 mark

Find the value of Q_3 for the following data set.

7 9 3 8 4 10 2 6 6 4

A. 4
B. 6
C. 8
D. 10

Edrolo U3&4 1E Q1

QUESTION 2 — 1 mark

The stem plot shows the *test scores* for a class of 23 students.

For this class, the interquartile range (IQR) of test scores is

A. 14.5
B. 17.5
C. 18
D. 24
E. 49

Key: $4 \mid 2 = 42 \quad n = 23$

4	0 1 4 4
5	2 7 9 9 9
6	5 6 8 8 9 9
7	0 0 5 6 7 8
8	5 9

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QUESTION 3 — 1 mark

The five-number summary for the amount 100 Year 12 students spent on games in the past year is: \$0, \$50, \$90, \$230, \$395.

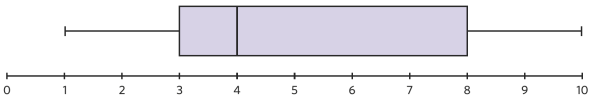
The lower fence for the data is

A. −\$220
B. −\$130
C. \$0
D. \$50

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QUESTION 4 — 1 mark

The boxplot is shown below.



The five-number summary for the data shown in the boxplot is

A. 1, 3, 4, 8, 10
B. 3, 3, 4, 8, 8
C. 1, 4, 4, 4, 10
D. 2, 3, 4, 8, 9

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QUESTION 5 — 1 mark

The table shows the mean and five-number summary of 800 recorded times, in milliseconds, between successive nerve impulses.

	time (milliseconds)
mean	220
minimum value	10
first quartile (Q_1)	70
median	150
third quartile (Q_3)	300
maximum value	1380

The shape of the distribution of these times is best described as

A. approximately symmetric.
B. positively skewed.
C. positively skewed with one or more outliers.
D. negatively skewed.
E. negatively skewed with one or more outliers.

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QUESTION 6 — 1 mark

Fill in the blanks:

For a negatively skewed boxplot, the median is towards the ____ of the ____. The right whisker will be ____ whereas the left whisker will be ____.

A. left, centre, short, longer
B. right, box, short, longer
C. middle, distribution, long, shorter
D. right, box, long, shorter

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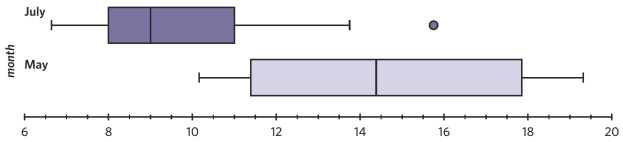
SECTION B · SHORT ANSWER

Show your working. 8 marks

QUESTION 7

2 marks

The boxplots show the distribution of maximum daily *temperature* (°C) for the months of May and July.



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a Determine the value of the upper fence for the July boxplot.

b Describe the shapes of the distributions of daily temperature (including outliers) for July and for May.

QUESTION 8

4 marks

The goal tally of the top 20 forwards in the 2000 AFL season is recorded.

51 47 52 70 57 53 56 76
42 53 50 109 49 43 47
44 42 69 40 68

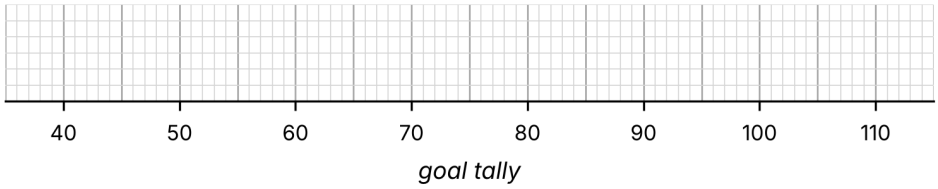
Edrolo U3&4 1E Q14

a Construct a five-number summary for the data.

b Calculate the IQR.

c Identify any outliers. Show working.

d Construct a boxplot to represent the data.



QUESTION 9

2 marks

Each dalmatian in a sample of 32 dalmatians had their *weight* recorded. The data is displayed in the ordered stem plot.

Key: 21 | 6 = 21.6 kg $n = 32$

21	6 9 9
22	1 2 5 6
23	0 1 4 6 6 7 8
24	4 5 6 7 7 9
25	6 8
26	1 7 9
27	3 7
28	2
29	1 8
30	4
31	1

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a Describe the shape of the distribution.

b Determine the median weight for this group of dalmatians.